



FACULTY	Agriculture, Engineering and Natural Sciences		
SCHOOL	Engineering and the Built Environment		
DEPARTMENT	Mechanical and Metallurgical Engineering		
SUBJECT	ENGINEERING MECHANICS II		
SUBJECT CODE	I3641NM		
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DURATION	2 Hours	MARKS	100

Second Opportunity Examination

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Internal Moderator: Dr. E. Shaanika

External Moderator: Prof. M Waheed (FUNAAB-NIGERIA)

This question paper consists of **4 pages** including this front page.

Instructions

1. Closed book examination
2. Read the questions carefully
3. This question paper consists of **Five** questions
4. Answer **any 4 Questions**
5. Each question carries **25 marks**
6. Neatly drawn sketches should be used where required
7. **Supplementary materials supplied: Appendix 1** – Center of Gravity and Mass Moment of Inertia of Homogeneous Solids

Question One

(a) An investigation was carried out to study the rectilinear kinematic motion of a particle. Illustrate the possible analysis you would likely consider to describe the characteristics of the motion if the acceleration is constant. **[8 Marks]**

(b) During a test, a rocket travels upward at 75 m/s, and when it is at position $S_A = 40 \text{ m}$ from the ground its engine fails. While in motion the rocket is subjected to a constant downward acceleration of 9.81 m/s^2 due to gravity. By neglecting the effect of air resistance. Determine:

- (i) the maximum height S_B reached by the rocket,
- (ii) its speed just before it hits the ground, and
- (iii) the total time of flight of the rocket just before touching the ground.

[17 Marks]

Question Two

(a) An investigation was carried out to study the curvilinear kinematic motion of a particle. Illustrate the possible analysis you would likely consider to describe the characteristics of the motion with respect to its rectangular components. **[10 Marks]**

(b) The velocity of a particle is given by $\mathbf{v} = \{16t^2\mathbf{i} - 4t^3\mathbf{j} + (5t + 2)\mathbf{k}\} \text{ m/s}$, where t is in seconds. If the particle is at the origin when $t = 0$. When $t = 2 \text{ s}$, determine:

- (i) the magnitude of the particle's velocity,
- (ii) the magnitude of the particle's acceleration, and
- (iii) the x, y, z coordinate position of the particle at this instant.

[15 Marks]

Question Three

(a) (I) Illustrate (i) a finite displacement of the motion of a particle acted upon by the force $\mathbf{F}$ along a curvilinear path from r_1 to r_2 or s_1 to s_2 ; (ii) motion of a particle under the action of a constant force $\mathbf{F}_c$ moving along a straight line; and (iii) an elastic spring elongated a distance ds . **[3 Marks]**
(II) Hence, demonstrate the work done by the forces in cases (i), (ii) and (iii) respectively of Q. 3 (a) with appropriate diagrams, and determine the work done by the respective forces.

[9 Marks]

(b) A 10-kg projectile is fired vertically upward from the ground, with an initial velocity of 50 m/s .

- (i) Determine the maximum height to which it will travel if atmospheric resistance is neglected, and
- (ii) Determine the initial velocity of the projectile after traveling 5 s from rest, assuming the air resistance is $F = 20t$.

[13 Marks]

Question Four

(a) Consider the angular motion of a body about a specified axis as shown in Figure Q. 4 (a). When a body rotates about a fixed axis, any point P located in the body travels along a circular path as indicated in the Figure. State:

- (i) appropriate relations of angular motion for a variable angular acceleration, and
- (ii) appropriate equations of angular motion for a constant angular acceleration.

[8 Marks]

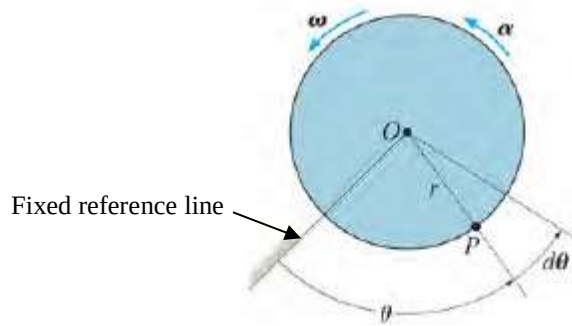


Figure Q. 4 (a)

(b) A pulley **A** attached to an electric motor is of radius 0.15 m and is used to turn a wheel **B** of radius 0.4 m . If the pulley **A** connected to the motor begins to rotate from rest with a constant angular acceleration of $\alpha_A = 2\text{ rad/s}^2$. After the pulley has turned 4 revolutions determine:

(i) the magnitude of the velocity, and (ii) the magnitude of the acceleration of a point at the edge of the wheel. Assume the transmission belt does not slip on the pulley and wheel.

[17 Marks]

Question Five

(a) The pendulum shown in Figure Q. 5 (a) consists of the 3-kg slender rod and the 5-kg thin plate.

(i) Determine the location $\bar{y}$ of the center of mass G of the pendulum, and

(ii) Calculate the moment of inertia of the pendulum about an axis perpendicular to the page and passing through G . Refer to **Appendix 1** for relevant values of mass moment of inertia.

[15 Marks]

(b) If the thin plate is centrally replaced with a slender rod of same mass as the plate, and of length 2 m ; determine:

(i) the new location $\bar{y}$ of the center of mass G of the pendulum, and

(ii) the moment of inertia passing through G .

[10 Marks]

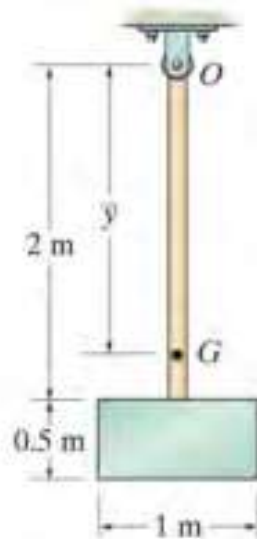


Figure Q. 5 (a)

Appendix 1: Center of Gravity and Mass Moment of Inertia of Homogeneous Solids

